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CEREBRAL PALSY AND CNS INFECTIONS

Cerebral Palsy (CP)

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INTRODUCTION :

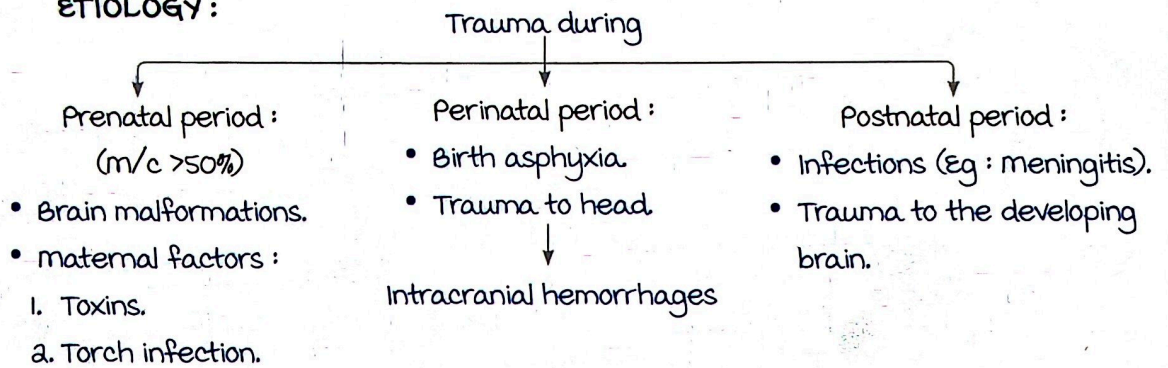
- AKA : Static Encephalopathy (Non progressive damage to developing brain).
- m/c chronic motor disability in childhood
- motor disabilities : Abnormality in

Tone

Posture

movement
- Intellectual disability : +/-
- manifestations : Evolve over a period of time.
- management : multidisciplinary.

ETIOLOGY :



CLASSIFICATION :

Based on motor manifestations.

1. Spastic CP :

- m/c type.

- Characterized by : ↑ tone/hyperreflexia.

Topographical Classification :

	Diplegia	Hemiplegia	Quadriplegia
Area of paralysis	Both lower limbs 	One side of the body 	All 4 limbs : • m/c in India. • m/c spastic CP. 
Seen in	Preterm babies	Neonates	Birth asphyxia of term babies
Sequelae of	Periventricular leukomalacia	Neonatal stroke	Parasagittal infarction

II. Dyskinetic/Choreo-athetoid CP :

- Dyskinesia : Abnormal movements of limbs.
 - Asphyxia / ↑ level of jaundice → Damage to Basal ganglia → Choreo athetoid/extrapyramidal movements.
- Kernicterus

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III. Ataxic CP :

- malformations in/toxin-damage to cerebellum vermis → Cerebellar dysfunction symptoms.

IV. mixed CP :

- Combination of all three.

CLINICAL PRESENTATION :

- Global Developmental Delay (GDD) : Delay in **at least 2 domains** of milestones (Gross motor, fine motor, language, social).
- Persistence of neonatal reflexes (As brain remains immature).
Eg : Moro's reflex : Persists after 6 months.
- Spasticity :
 - Scissoring posture.** ← 
 - Seen in spastic diplegia type of CP.
 - D/t ↑ tone of adductors in lower limb.
 - Toe walking : Involuntary.
 - Cortical thumb : D/t ↑ tone of thumb.** ← 
 - Thumb in a fist posture.
 - Abnormal : If (+) > 4 months.
- Vision abnormalities.
- Hearing abnormalities.

MANAGEMENT :

IOC : **MRI Brain** (To localize area of damage).

Rx : Treat symptomatically with multidisciplinary approach d/t irreversible nature.

- Seizure → Anti epileptics.
- Spasticity : muscle relaxants : Baclofen, Botulinum toxin A.
- Contractures → Tendon release procedure.
- Dystonia/Dyskinesia → Trihexyphenidyl, Levodopa.

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CNS Infections

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PYOGENIC / BACTERIAL MENINGITIS**Etiology :**

- 0-2 months : *E. coli*.
- 2-24 months : *H. influenza* > *Strept. pneumoniae*.
- >2 years : *Strep. pneumoniae* (Overall m/c cause).

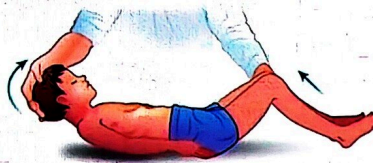
Clinical features :

- Fever.
- Others : GTCS seizures, irritability.
- Signs of ↑ ICP (Intra-cranial pressure)
 - Infant: Bulged anterior fontanelle
 - Older child
 - Projectile (Uncontrolled) vomiting
 - Occipital headache
 - Visual defects : Papilledema
- Signs of meningeal irritation (Uncommon < 2yrs).

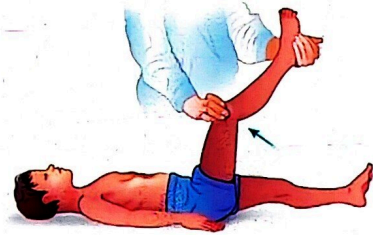
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**1. Neck stiffness / Nuchal rigidity**

- Painful flexion at neck.
- Keeps the neck stiff.

**2. Brudzinksi's sign**

Flexion of hips and knees in response to neck flexion

**3. Kernig's sign**

Resistance to extension of leg while the hip is flexed

- Flex hip & knee
- ↓
- Extend the knee
- ↓

• Pain in the spine region
(D/t stretching of inflamed meninges)

Investigations : CSF analysis :**Lumbar puncture (IOC) :**

- C/I : Severe signs of ↑ ICP → Instead → Contrast enhanced CT (CECT) : Shows meningeal enhancement.
- ↓
- Can lead to herniation
- ↓
- Death of child

Findings :

CSF parameters	Findings
Appearance of CSF	Turbid/Cloudy
Inflammatory cell	↑ (Abundant neutrophils)
Protein	↑
Sugars	↓ (Hypoglycorrhachia)

(Normal CSF : <5 lymphocytes/mm³)

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Gram's staining.
Culture & sensitivity (C&S).
PCR : more sensitive.

} To detect causative bacteria

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Complications :

Acute :

- m/c : Seizures.
- Brain abscess.
- Subdural effusion.

Long term/chronic :

- m/c : Sensorineural hearing loss (SNHL).

Particularly in *S. pneumoniae* associated meningitis.

Brain evoked response audiometry (BERA).

Treatment :

Antibiotics :

- 3rd generation cephalosporins : Ceftriaxone/Cefotaxime.

↓ No response in 48 hours

Add Vancomycin

- Can change antibiotic post C&S results.

• Duration of Rx :

- Child : 7-10 days.
- Neonate : 3 weeks.

Steroids : Along with first dose of IV antibiotics.

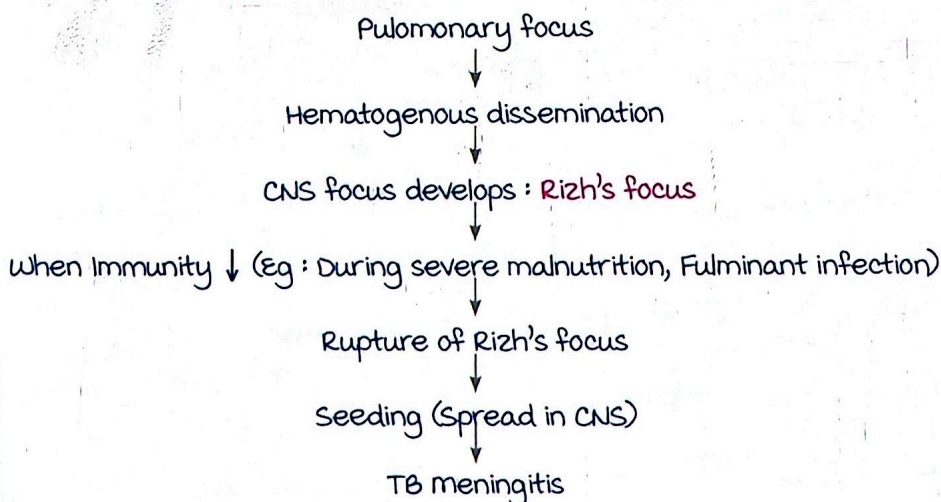
- DOC : Dexamethasone x 2 days.
- Goal : ↓ inflammation, ↓ incidence of SNHL.

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TB MENINGITIS :

- most severe form of TB in children.

Pathophysiology :



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Clinical Features :

Prodromal phase : 1-4 weeks

- Fever/ Irritability/ Vomiting.
- Not conclusive of TB meningitis.

If
undiagnosed/
untreated

meningitis

- Features of :
 - meningeal irritation.
 - Photophobia.
 - Seizures.

If undiagnosed/
untreated

Coma :

- ↑ mortality.

Diagnosis :

1. CT Brain :

- **Thick exudates in basal cistern** : Characteristics feature.

2. CSF Findings :

- may mimic bacterial meningitis/normal 10-15% cases.
- Cell : Lymphocyte predominance.
- **Protein : very high** ($> 1000 \text{ mg/dL}$) {Normal : $20-45 \text{ mg/dL}$ }.
- **ADA** (Adenosine deaminase) : ↑.
- **Ziehl Neelsen staining** : Identifies TB bacilli.
- **CB-NAAT** (Cartridge Based - Nucleic acid Amplification Testing).
- **Cobweb appearance of CSF**.
 - When CSF is left undisturbed in a test tube.

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Treatment :

Anti-tubercular treatment : 9-12 months

I. Intensive Phase (2 months)

- **HRZE** regimen
 - H: Isoniazid
 - R: Rifampicin
 - Z: Pyrazinamide
 - E: Ethambutol

II. maintenance Phase (7-10 months :

Depending on clinical status of patient)

- **HR** regimen.

Steroids : For 6-8 weeks.

- IV Dexamethasone → Oral Steroids.
(Few days)

VIRAL MENINGOENCEPHALITIS :

Encephalitis : Cortical lobe involvement

→ Signs of involved lobe (Eg : Speech defect, focal seizures).

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Etiology :

- m/c worldwide : Japanese Encephalitis virus.
- m/c cause of sporadic encephalitis : HSV₁ (Herpes Simplex Virus).

Japanese Encephalitis (JE) :

- vector : Culex tritaeniorhynchus (Culex mosquito).
- Seen in rural population (Associated with rice fields).
- Host : Ticks (Amplifier host), birds.
- Age : School-going child.

Diagnosis :

- PCR : Specific test.
- CSF analysis
 - Cell count : ↑ Lymphocyte
 - Sugars : Normal
 - Protein : Slight ↑
- Imaging.
 - Target areas : Basal ganglia > Thalamus.



Basal ganglia and thalami

Management :

- Supportive treatment.
- Prevention : JE vaccine.

Herpes Simplex Encephalitis :

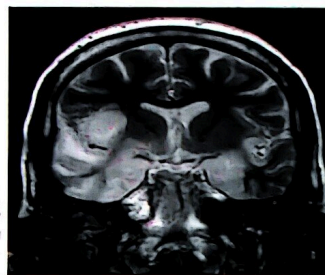
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Features :

- Fever.
- Focal seizures.
- Focal neurological deficit (Eg : Aphasia, speech defects).
- Behavioral disturbances.
- movement disorders.

Investigation :

- CSF Analysis
 - Cell count : ↑ Lymphocyte
 - Sugars : Normal
 - Protein : Slight ↑
- PCR.
- Imaging.



Temporal lobe involvement

- Target areas : Temporal lobe.
- EEG
 - PLED (Periodic Lateralizing Epileptiform Discharges) : Characteristic finding.

Treatment :

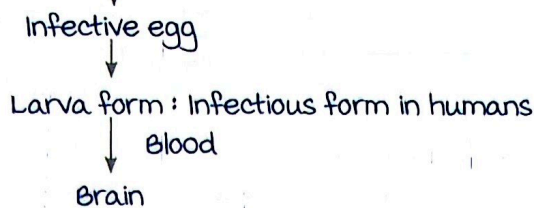
DOC : IV Acyclovir x 14-21 days.

NEUROCYSTICERCOSIS :

Causative Agent : (Taenia solium) (Pork tapeworm).

Age : usually affects children > 3 years.

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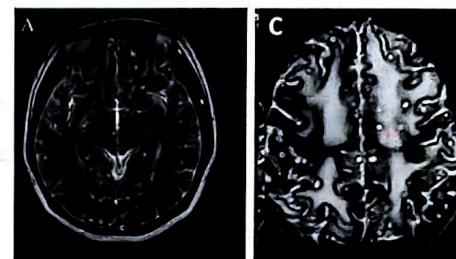
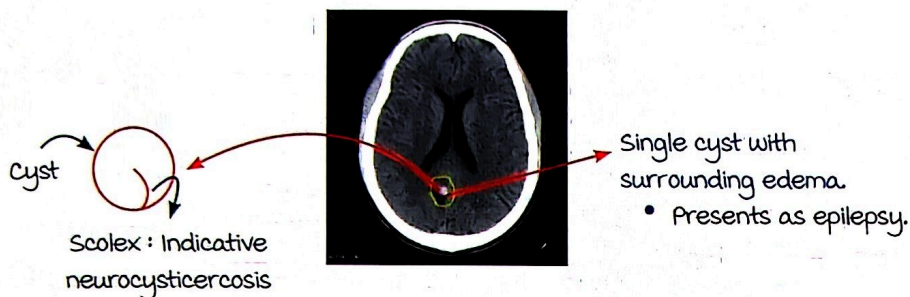
Clinical presentation :

Encephalitis (Acute presentation).

- Fever.
- Altered sensorium.
- Seizures.
 - H/O recent onset focal/generalized epilepsy.
 - Seen if cysts are less in number.

Investigation : Imaging of Brain.

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Treatment :

Steroids :

- To reduce inflammation.
- DOC : Dexamethasone.

↓ 48 hours

Cysticidal therapy :

- Single cyst : Albendazole (10-14 days).
- Multiple cysts + : Praziquantel.

Anti-epileptics (If seizures +) : For 6 months

Focal epilepsy :
Oxcarbazepine

General epilepsy :
Sodium valproate